

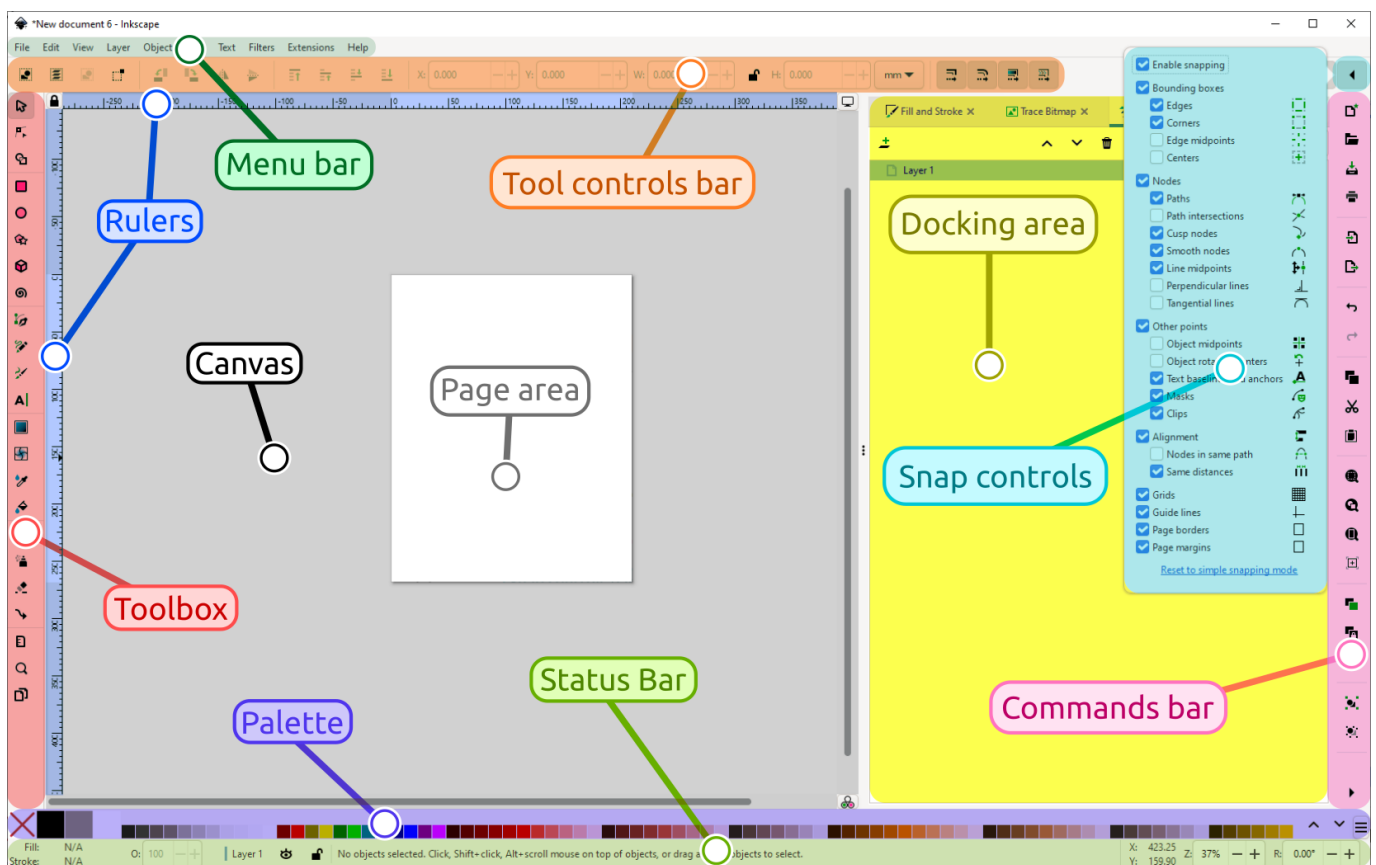
Introduction

In this exercise, you will design your own wooden tea light candle holder, which will later be milled from a piece of selected hardwood. The aim is to introduce you to the basics of Inkscape and show you how to create simple vector shapes using Bézier curves. Your task is simple: draw the outline of your desired candle holder and place a specified-diameter circle in its centre.

This sketch will then be imported into CAM software, where you can see how the toolpaths are programmed for cutting out your design from the wooden block and machining the pocket to hold the candle.

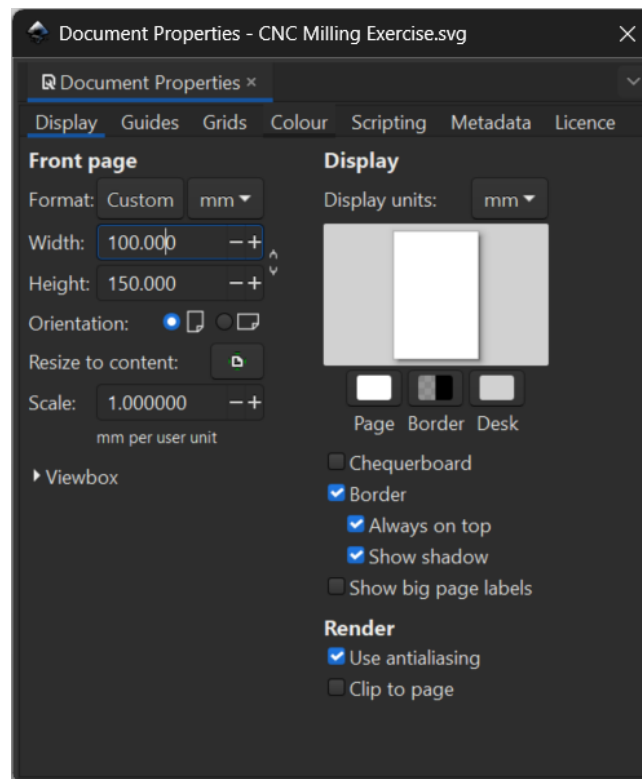
For inspiration, we have prepared a Christmas-themed example: a Christmas tree-shaped candle holder. This manual contains all the information you need to set up the Inkscape document, as well as explanations of how to use Inkscape's tools to draw, edit and align objects on your page.

Document preparation



Setting the document page size, preferences and orientation

Launch Inkscape and select **New Document**. Next, in the menu bar, navigate to **File -> Document Properties...** (or use **Shft+Ctrl+D**). Set the display units to millimetres and the page size to 100 by 150 mm. Depending on your design, you can change the page orientation to landscape.



Next, in the menu bar, go to **Edit -> Preferences** . In the **Preferences** window, navigate to **Tools** and select the **Geometric bounding box** under the **Bounding box to use** .

Starting your sketch

Using the Pencil tool

Select the **Pencil** tool from the **Toolbox** or press the **P** button on your keyboard.

New options will then appear in the tool controls bar.



The **Mode** will be set to **Create regular Bezier path** by default.

Smoothing is the process by which many small, jagged input points are converted into fewer, cleaner Bézier segments. This setting controls how aggressively the tool simplifies and fits curves: higher smoothing reduces noisy points and produces gentler curves, while lower smoothing preserves detail. Start by setting the smoothing value to 45, then adjust it to your liking.

The Pencil tool lets you draw paths by placing and connecting nodes with simple mouse actions. **To draw freehand, press and hold the left mouse button and drag** — the tool will record your mouse movements and create a continuous Bézier path. To create a straight-line path, click to place a node, move the pointer and click again to place the next node — each click adds a

straight segment between nodes. Releasing the button without dragging will prevent curve handles from being added. To connect nodes, click the end node of one path and then the start node of another path (or click the first node of the same path) to close the path.

Using the Node tool

You can find the **Node tool** in the **Toolbox** (second from the top), or you can enable it by pressing the **N** key on your keyboard. The Node tool allows you to refine any path after it has been drawn. When you select a path with the Node tool, all its nodes (anchors) and their Bézier handles become visible. You can then move the nodes to change the overall shape or drag the handles to adjust the direction and strength of the curve.

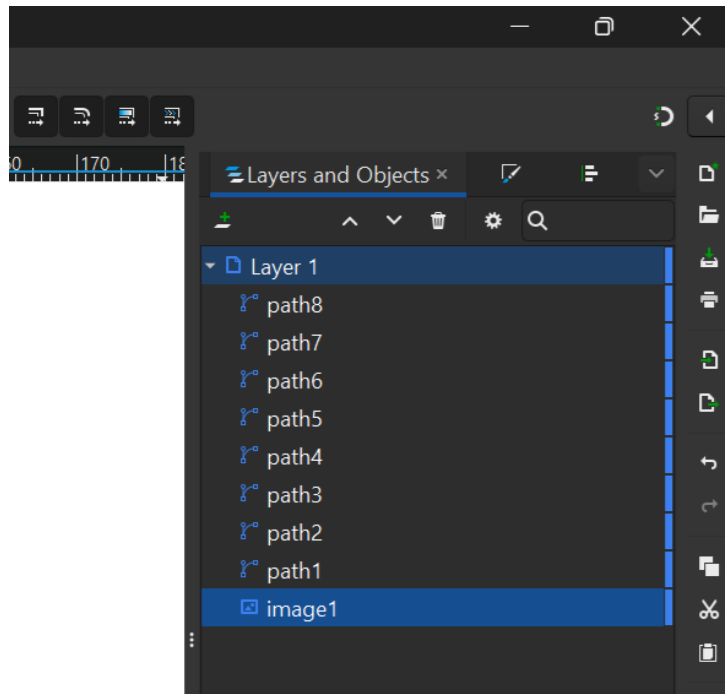


Once again, options specific to the selected tool will appear in the **controls bar**. Here, you can add or delete nodes to simplify a shape or break a node's handles to create a cusp. The Node tool also lets you join or separate paths, close open shapes and convert between node types. Several node types are available in the control bar: smooth nodes, which create continuous, flowing curves; symmetrical nodes, where both handles are the same length; and corner nodes, which allow for sharp angles or independently controlled handles. Hover your mouse over the options in the controls bar to see what they are.

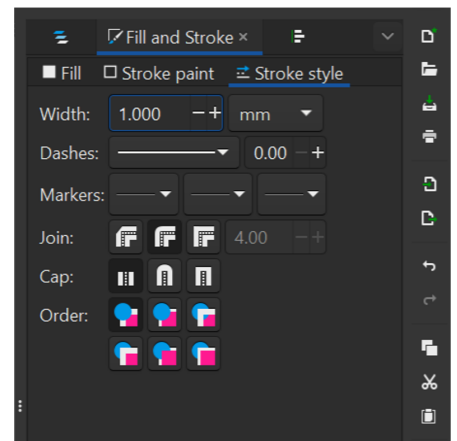
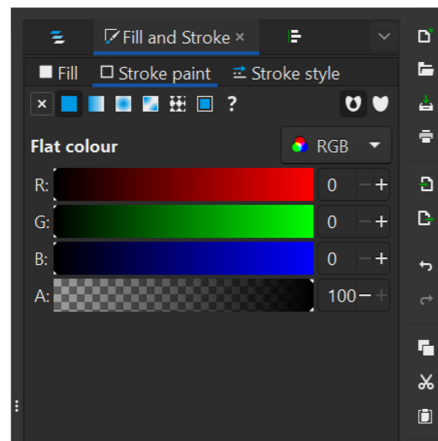
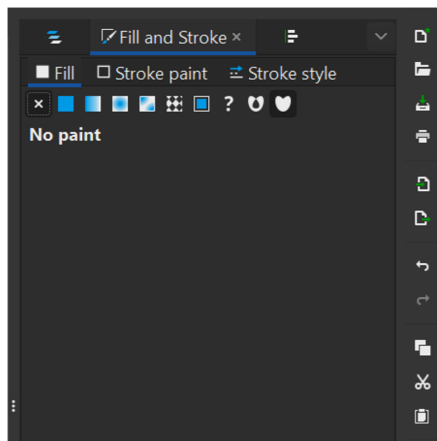
Creating your first drawing

You can use the pencil tool to draw the desired shape freehand or trace it from an image. This can be a picture from the internet or a sketch you made on a piece of paper. Whatever your reference picture, you can insert it into the page by going to **File -> Import**. Use the **Selector Tool**, found at the top of the **Toolbox**, to move and scale your picture, or simply press **S** on your keyboard. To keep the proportions of your picture when scaling it, make sure that you lock the dimensions by clicking the **padlock icon** visible on the right of the **Selector Tool controls bar**.

To make sure we can draw on top of our picture, select **Object -> Layers and Objects...** from the menu bar (or use Shift+Ctrl+L). In the docking bar you will now see all of the layers and objects in your document. Click on your picture and drag it to the bottom of the list.



If you want to change the colour or the line thickness of drawings made with the pencil tool (for instance to make them more visible on top of a picture you inserted), in the menu bar navigate to **Object -> Fill and Stroke...** (or use Shift+Ctrl+D). This will open **Fill and Stroke** tab in the docking area. There, you can select the fill, stroke and stroke style of objects selected with the selector tool.



For the purposes of this exercise, make sure to set **Fill** to **No Paint** and then set the **Stroke paint** to **Flat colour** and pick the desired colour from the sliders. In the **Stroke style** set the line **width** according to your preference.

Making the tea light candle hole and aligning it on the page

From the **tool box**, select **Ellipse/Arc Tool**. Hold the **Control** key on your keyboard and drag your mouse while holding the left mouse button to draw a circle. Then select your circle with the selector tool and set its width and height to 39.5 mm.

You can then position the circle by simply dragging it into a desired place or use **Align and Distribute tool** to position it precisely in the centre relative to your drawing or page. This tool can be found in the **menu bar** under **Object -> Align and Distribute**